

consumed visual block

tokens the LM reads, h_{it}
(final block; pooled h_i)

read

logistic probe on h_i
→ direction $\hat{w}_c$
+20 random-label controls

write

$h' = h + \alpha \|h_t\| \hat{w}_c$
same tokens, $\alpha = +0.25$

answer

$P(\text{yes} | q)$, fp32 logits
 $W_{q,d}$: paired change

compare on the same rows, dose, and endpoint

- concept write $W_{q,q}$
- five competing clinical writes $\max_{d \neq q} W_{q,d}$
- 119 random directions 95th percentile
- coordinate-permutation sham $|\text{sham}|$

owned: $O_q = W_{q,q} - \max_{d \neq q} W_{q,d} > 0$ (max- T bounds), $W_{q,q} > \text{p95}$, $|\text{sham}|$

